

System Requirements Document (SRD)

Example Autonomous Warehouse Inspection System

Document Control	Value
Document ID	SRD-AWIS-001
Version	1.0
Status	Example / Training Reference
Input	Approved BRD-AWIS-001
Purpose	Define WITH WHAT the solution will be realized and HOW MUCH / HOW MANY capability each selected system element must provide.

1. System Requirement Method

Each SRD requirement selects or constrains a major system resource and attaches measurable specifications. Quantities and performance values are derived from BRD outcomes rather than selected arbitrarily.

2. System Context

The system comprises indoor autonomous mobile robots, onboard navigation/perception, AI inference, wireless communications, fleet services, operational data storage, and browser-based operator functions. Detailed interaction belongs in the SDD.

3. Key Derivation Assumptions

- Example warehouse area: 50,000 m² with 20 km aggregate inspection route.
- Complete inspection cycle required <=4 hours (BR-T01).
- Allowance required for charging, congestion, recovery and maintenance reserve.
- Final quantities must be validated by route simulation and pilot measurement.

4. Selected System Elements - With What / How Much / How Many

SR ID	System Element / With What	Measurable Requirement
SR-001	TurtleBot4-class AMR platform (example selection)	12 operational + 2 spare/site; inspection speed ≥ 0.5 m/s; usable runtime ≥ 8 hr/charge.
SR-002	Primary 2D ranging sensor	1/robot; ≥ 10 Hz; usable range ≥ 12 m; 360° horizontal coverage.
SR-003	OAK-D Pro-class RGB-D camera	1/robot; RGB $\geq 1280 \times 720$ @ ≥ 30 FPS; usable depth 0.4-5 m.
SR-004	Onboard compute platform	1/robot; navigation + AI workload with average CPU/GPU utilization $\leq 80\%$.
SR-005	Ubuntu 24.04 LTS	Common approved OS/patch baseline on robot compute and designated edge servers.
SR-006	ROS 2 Jazzy	DDS communications; namespace architecture supports ≥ 20 robots/site.
SR-007	Nav2 navigation framework	Controller ≥ 20 Hz; global planning ≥ 2 Hz under nominal acceptance load.
SR-008	AMCL-compatible localization	95th percentile position error ≤ 0.10 m; heading error $\leq 5^\circ$.
SR-009	Validated YOLO-family detection model	Inference ≥ 20 FPS and capable of BR-T02 precision/recall targets.
SR-010	Fleet management server	Support ≥ 20 connected robots and $\geq 1,000$ missions/day/site.

5. Infrastructure, Data and Interface Requirements

SR ID	System Element / With What	Measurable Requirement
SR-011	Managed Wi-Fi 6/6E	Coverage ≥ -67 dBm on routes; packet loss $< 1\%$; robot-service RTT ≤ 50 ms p95.
SR-012	PostgreSQL operational database	Retain ≥ 12 months online; ≥ 50 M records; daily backup; RPO ≤ 24 hr.
SR-013	Web operator application	Support ≥ 100 concurrent authenticated users/site; response ≤ 2 sec p95.
SR-014	Notification service	Critical alert delivery initiation ≤ 2 sec after receipt, supporting ≤ 5 sec end-to-end.
SR-015	Charging infrastructure	At least 1 charger/4 operational robots + maintenance charger; 20%-80% ≤ 2.5 hr.
SR-016	Central logging/monitoring	Searchable logs ≥ 90 days; critical infrastructure alarm ≤ 60 sec.
SR-017	Configuration/version repository	100% deployed versions traceable; current + previous release rollback retained.

6. Interface Requirements

Interface	Requirement
Robot-Fleet	Authenticated mission/status exchange; ≥ 1 Hz status update/robot.
Perception-AI	Timestamped image/depth; frame and calibration metadata preserved.
Fleet-Dashboard	Versioned API; no direct dashboard dependency on internal robot nodes.
System-WMS/ERP	Initial release uses approved read-only inventory/location master-data interface.

7. Safety, Reliability, Security and Maintainability

SR ID	Category	Requirement
SR-020	Safety	Physical emergency stop; stop function shall not depend on cloud/dashboard availability.
SR-021	Safety	Collision protection shall use an independently justified sensing/control path; AI alone is not sole protection.
SR-022	Safe failure	Loss of localization, commanded-motion control, or required safety sensor causes defined safe stop.
SR-023	Availability	One robot unavailable shall not prevent BR-T01 cycle under nominal conditions.
SR-024	Reliability	Automatic recovery for defined transient faults; max 3 attempts before escalation.
SR-025	Cybersecurity	Named authenticated accounts, RBAC, audited administrative actions.
SR-026	Cybersecurity	Control, credential, and business-data communications use approved encryption.
SR-027	Maintainability	Field-replaceable sensor/compute restored to approved config <=60 min.
SR-028	Software maintenance	Remote deployment, health verification, and rollback supported.

8. Environmental and Physical Constraints

Item	Requirement
Environment	Indoor; 5-35°C; 20-80% RH non-condensing.
Floor	Dry, level industrial floor within platform limits.
Lighting	>=150 lux unless supplemental illumination provided.
Clearance	Robot envelope + safety clearance fits approved aisle geometry.

9. Verification and Acceptance Matrix

SR / BR Link	Verification Method	Example Acceptance Evidence
SR-001 -> BR-T01	Analysis + simulation + pilot	Fleet demonstrates complete route cycle <=4 hr with reserve allowance.
SR-003/SR-009 -> BR-T02	Controlled validation	Precision >=95%, recall >=90%; live inference >=20 FPS.
SR-011/SR-014 -> BR-T05	Network/load + end-to-end	Critical notification <=5 sec p95 under acceptance load.
SR-008 -> BR-T06	Surveyed route test	Localization <=0.10 m / <=5° p95 and mission success target.
SR-012 -> BR-T07	Inspection + restore	Capacity, query, backup and restore evidence.
SR-023 -> BR-T08	Analysis + endurance	Capacity and trial support >=99.5% scheduled availability.

10. Example Requirement Derivation

BR-T01 requires 100% inspection coverage every four hours. A 20 km aggregate route at 0.5 m/s represents about 11.1 hours of pure travel for one robot before inspection dwell, charging, congestion, recovery, and maintenance reserve. Parallelization therefore drives fleet quantity. The example selection of 12 operational plus 2 spare robots must be validated by simulation and pilot measurement.

11. Boundary to the SDD

The SRD selects/constrains system resources and measurable capability. The SDD explains HOW these elements interact: architecture, ROS node allocation, interfaces/topics/APIs, data flow, state machines, deployment, timing, error handling, recovery, cybersecurity controls, and safety mechanisms.

12. Glossary

SRD: System Requirements Document. SDD: System Design Document. RTT: Round-Trip Time. RPO: Recovery Point Objective. AMR: Autonomous Mobile Robot. DDS: Data Distribution Service.